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RESEARCH ARTICLE

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DCapsNet: a robust multi-feature deep capsule network for speech emotion recognition

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ABSTRACT

With the continuous advancements in artificial intelligence, speech emotion recognition (SER) systems have garnered widespread attention. They have found applications across various fields, including mobile services, psychological assessment, healthcare, and autonomous vehicles. This research introduces a novel multi-feature deep capsule network capable of real-time speech processing. Firstly, speech preprocessing and data enhancement techniques were used to enhance the dataset, followed by the extraction of discriminative features. To construct the SER model, spectral and temporal information was extracted as a set using three datasets: SAVEE, RAVDESS, and EMO-DB. These feature sets were given to the deep capsule network (DCapsNet). The capsule layers in DCapsNet preserved the spatial relationships within speech signals and ensured robust feature representation, allowing for accurate emotion recognition. The model employed a hierarchical learning using aggregated capsule outputs to enhance recognition performance. Finally, the performance of DCapsNet is assessed and compared with that of the baseline models using the multilingual speech datasets. The results demonstrated that the DCapsNet achieved superior performance compared to existing methods in the field. The micro accuracy of DCapsNet for the RAVDESS, SAVEE, and EMO-DB datasets is 94%, 90%, and 93%, respectively. This study demonstrated the effectiveness of the DCapsNet for recognizing speech emotions.

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KEYWORDS
SER; feature fusion; data augmentation; segmentation; deep capsule network

1. Introduction

Speech is an effective and natural communication method and carries both linguistic and paralinguistic characteristics such as accent, age, gender, and emotions [1]. Emotions can be identified through various modalities, including facial expressions, physiological signals, and speech [2, 3]. Speech emotion recognition (SER) is a process used to identify individual emotions from a known set of audio data based on voice patterns [4]. SER is a crucial aspect of human-computer interaction (HCI), enabling computers to respond more empathetically to human emotions. The identification of emotions in human communication has played a crucial role in the development of HCI-based applications, as shown in Figure 1.

Significant research has been reported in the field of SER for the continuous refinement of machine learning (ML) algorithms. These algorithms aim to recognize emotional cues from input data and enable machines to understand and respond appropriately to human emotions [5]. Recently, SER systems have been widely applied in various applications, including mental health monitoring, which involves identifying different psychological and behavioural symptoms that can have a substantial impact on various aspects of life. Mental health challenges such as anxiety, stress, and depression can lead to disability and are common among individuals with chronic diseases [6]. Despite the severe consequences of mental health issues, regular monitoring remains limited due to high costs [7]. Speech signal processing can help to recognize such patterns. For instance, call centers and voice portals use anger monitoring tools [8] to detect emotional state accurately which helps to enhance overall customer satisfaction. Moreover, civil aviation has started monitoring pilot stress levels [9]. SER systems can reduce the risk of potential aircraft accidents by catching high-stress levels. These systems are also applied in Parkinson's disease detection [10], pain assessment, post-traumatic stress disorder [11], and Alzheimer's disease identification [12]. The recent advancement in SER technology has offers several benefits. However, it has also arise ethical concerns with its development and application. For instance, the potential misuse of personal data. SER systems often rely on data that include sensitive and emotional patterns. This data could lead to breaches of privacy. Therefore, it is essential to implement data protection protocols and enforce regulations to mitigate these risks.

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Figure 1. Applications of SER.

Existing SER systems have focussed on mining phonetic features from speech to accurately transcribe uttered words [13]. However, such systems often neglect key para-linguistic cues present in speech, such as mental state, gender, and intent. The human brain instinctively combines both phonetic and para-linguistic information to perceive the meaning of utterances. Para-linguistic cues play a crucial role as they provide additional context and depth. In contrast, SER systems are focussed on phonetic elements and leave para-linguistic information aside. This limited focus has hindered the ability of SER technology to retrieve the complexity of human conversation. Recent research in machine learning (ML) has opened new avenues for integrating para-linguistic information into SER systems. Previously, authors have been particularly interested in emotion recognition. Studies in this area focussed on emotions detection through facial expressions. Nonetheless, significant progress has been made in SER in recent years. Emotion recognition is vital for improving HCI. Therefore, SER systems have gained popularity for analyzing emotions based on vocal cues. Despite this growing focus, SER remains a challenging task due to the complexity of features. Thus, the development of advanced feature extraction techniques in SER remains a critical area of research.

Recent research has proposed several approaches to enhance the diversity and performance of SER systems. These approaches involve the fusion of different characteristics derived from audio files of each dataset. The primary objective of these characteristics is to improve the robustness, and generalization of SER models. Moreover, various feature extraction methods are employed to minimize uncertainty and enhance the accuracy. These methods aim to enhance the effectiveness of SER systems by using a diverse range of robust features. This study derived both time and frequency-based features from each dataset file for SER. The RAVDESS, SAVEE, and EMO-DB datasets, which are commonly used in the development of SER systems, were chosen for this study. These features were then fed into an ML model. For evaluation, the ML model demonstrated improved recognition rates for each SER model. The contributions are:

- Data augmentation is applied to enhance the training dataset and improve the model generalization.
- This study combines several acoustic descriptors to create a unified and rich emotional representation. The capsule layers used these fused characteristics to extract both local (segment-level) and global (contextual) dependencies.
- A one-dimensional capsule network is specifically designed for SER. The network incorporates convolutional layers for low-level feature extraction. The primary and lambda capsule layers used dynamic routing-by-agreement for hierarchical relationships between features and emotions.
- Extensive experiments on benchmark datasets demonstrated that DCapsNet achieved consistent improvements and robustness over traditional models.

The rest of paper is organized as follows: Section 2 presents a comprehensive review on SER techniques. Section 3 describes the proposed methodology, and the experimental findings are discussed in Section 4. The significance of the various experimental results is analyzed in Section 5. Lastly, Section 6 concludes the article with potential avenues for future research, inspiring further exploration and innovation in the field of SER.

2. Literature review

Significant research has been carried out in SER using speech data. This data is recorded through various devices in either real-time or controlled environments. Recently, numerous studies have been published on SER. These studies employed both conventional ML classifiers and DL techniques to identify speaker emotions. A study [14] extracted traditional features from the EMO-DB dataset. These features were given to k-nearest neighbour (k-NN) and support vector machine (SVM) classifiers to recognize the emotions. Recent studies employed pre-trained models such as wav2vec 2.0 [15] and HuBERT [16] on large-scale corpora and provided rich contextualized representations. These self-supervised models [17] have shown significant improvements in emotion recognition performance across multiple benchmark datasets. However, existing SER systems still face several key limitations. Many ML approaches depend heavily on manual features which often fail to generalize across speakers, languages, and recording environments. Similarly, DL models frequently require large annotated datasets and may overfit on small or imbalanced data. These challenges highlight the need generalized SER architectures

capable of capturing both local and global emotional dependencies in speech.

In the field of SER, several DL techniques like LSTM, RNN, CNN, and their combination have been studied. Particularly, CNN variants have been investigated to recognize emotions by extracting suitable information from time-series speech data. For instance, [18] proposed a deep CNN for SER utilizing three open datasets, i.e. IEMOCAP, K-EmoCon, and SEWA. This network used the temporal dependencies of the time series to capture the suitable features. To select the best parameters for CNN, Agrawal and Jain [19] reported a 1D CNN for SER and conducted a thorough investigation for a metaheuristic approach. The authors utilized the RAVDESS, SAVEE, and novel ANAKE Hindi speech datasets to evaluate the algorithm's optimization. The results achieved revealed the advantage of the metaheuristic algorithm (Bat Random Forest Hybrid Meta Heuristic Algorithm) to effectively optimize CNN parameters. Similarly, authors of the study [20] employed a multi-learning trick (MLT) based on a 1D CNN to retrieve global and local characteristics from voice signals. The proposed method retrieved the robust and suitable features using a dynamic fusion model to enhance the recognition rate. The developed MLT model was evaluated on both long and short-term relative dependencies using EMO-DB and IEMOCAP datasets. The proposed method obtained an average accuracy of 90% and 73%, respectively, on each dataset. Nonetheless, the model training and testing times were relatively high for real-time speech compared to baseline models. Although CNNs effectively capture local spectral-temporal patterns, they often fail to retain long-term dependencies essential for recognizing emotions that evolve gradually over time.

RNN is explicitly designed to retrieve temporal dependencies in time series data. Therefore, several RNN variants, such as LSTM and BiLSTM, have been implemented to address the challenges of SER [21]. The study [22] reported different variants of RNN to recognize the speech emotions. The proposed approach utilized both unidirectional and bidirectional RNNs for SER applications. Similarly, Barhoumi and BenAyed [23] investigated a Bi-LSTM using time series data collected from a mobile device in a real-time environment. The optimum accuracy obtained by the Bi-LSTM model using different acoustic features was 98.51% and 86.85% on the EMO-DB and RAVDESS datasets, respectively. To effectively address the overfitting concerns, the study [24] proposed an approach for reducing feature dimensionality. The authors first introduced a novel method for SER that derived multiple short- and rhythmic characteristics, i.e. zero-cross rate, short-time energy, and mean amplitude difference. After the feature extraction phase, an LSTM network was employed for experiments that achieved impressive performance of 98.5%, 100%, and 98.9% on CASIA, Emo-DB, and EMOVO datasets, respectively. Although the proposed approach demonstrated its capability to correctly recognize speech emotions across English and German languages, it has not been evaluated on low-resource languages like Urdu and Hindi. Moreover, the sample size used is very small, and the model is not robust or generalized enough to be applied in complex and real-time environments. In addition to dimensionality reduction, other effective approaches to mitigate overfitting in SER include transfer learning [25, 26] and adversarial learning [27]. Transfer learning enables leveraging knowledge from large-scale pre-trained models, making it especially useful for small emotional corpora. In contrast, adversarial training techniques enhance the robustness to noise and distribution shifts. However, these techniques often struggle with vanishing gradients and long training times which limit their scalability for real-world applications.

Khan et al. [28] designed a robust and reliable multimodal SER system to identify emotions through multi-modality. The study proposes a novel feature fusion method that utilizes a cross-attention mechanism. The voice signals and text were processed by the CNN to extract semantic and discriminative features. The multi-headed cross-attention mechanism was used to enhance the interaction between speech and text by cross-attention to retrieve relevant and suitable characteristics for emotion detection. Lastly, the region-wise weights of both encoders were combined to enable the interaction among various layers and paths for feature fusion. The proposed approach was evaluated using MELD and IEMOCAP datasets and achieved significant improvement over baseline methods. Despite these advancements, most multimodal and transformer-based methods rely on large, high-quality labelled datasets and high computational resources, making their deployment challenging in practical SER scenarios. To evaluate the efficacy and improve the accuracy of SER systems in cross-corpus tasks, Ahn et al. [29] investigated two tasks: the multitask framework and contrastive learning. For better understanding, the secondary task utilized the target data without class labels. The multitask framework employed a pre-trained transformer model. The multitask transformers model was developed for text and speech and conducted decision-level integration during inference. Again, this study also utilized public datasets, including EMO-DB, MSP-IMPROV, and IEMOCAP. The results revealed that improved performance was achieved on these datasets in cross-corpus settings.

In another study [30], the authors utilized the RECOLA and IEMOCAP datasets to develop a deep neural network (DNN) model for SER. The study revealed that the performance of the SER system was optimum when the proposed model was utilized with context stacking and root mean square aggregation. The DiCCOSER-CS enhanced the valence CCC by

12.6% and the arousal CCC by 9.6% in regression when compared with the hybrid CNN-LSTM. Zaheer et al. [31] used a variant 5-layered ML algorithm on the SEMOUR dataset and gained an accuracy of 90%. However, no DL based technique was employed on SEMOUR for speech emotion detection. An RNN model has been employed with minor variations for feature extraction in SER applications. For this, Mirsamadi et al. [32] implemented a deep RNN to retrieve the frame-level categorization along with temporal set into long intervals. The study also reported a weighted time-pooling method to identify emotion-prominent fragments via an attention mechanism. The proposed approach achieved 3.1% unweighted and 5.7% weighted accuracy compared to the conventional SVM-SER approach using the IEMOCAP dataset. Although the proposed approach presented a novel algorithm for SER, the experiments were conducted using only one balanced dataset. To handle this, Ai et al. [33] investigated an attention-based convolutional RNN. The results on IEMOCAP and Emo-DB demonstrated superior performance for imbalanced audio files. In another study, a self-attention mechanism [34] was introduced to the BiLSTM. The proposed approach effectively captured correlations within speech signals and enhanced the representation of diverse information by utilizing directional self-attention (DSA). The method demonstrated strong performance when evaluated on both the IEMOCAP and the EMO-DB datasets.

Transformer-based models typically require high computational resources to capture long-term dependencies. A related work named MemoCMT MemoCMT [35] addresses this issue and introduced a cross-modal transformer capable of integrating both local and global speech-text features. The classification was performed on fused features that were obtained using BERT and HuBERT models and the model achieved notable results on IEMOCAP and ESD. In a similar work, Khan et al. [36] researchers proposed the Joint Multi-Scale Multimodal Transformer (JMMT) model which studied the combined relationship among multiple modalities. This model established a joint representation through multi-scale and recursive cross-attention mechanisms and gained improved results over IEMOCAP and MELD datasets. Besides the cross-attention methods, researchers have focussed on adding ASR-generated transcripts to SER models. In one such study, the authors [37, 38] determined how the word error rate and several fusion mechanisms affected the emotion recognition performance. Another similar study [39] combined audio and text at an early stage of the proposed model, whereas [40] introduced a dual-memory fusion method that helped the model learn long-term dependencies between speech and text data more effectively.

The aforementioned studies have significantly improved SER systems through assorted feature extraction methods and learning model architectures. However, current systems still face challenges in achieving robustness, generalization, and computational efficiency simultaneously. Specifically, only a few studies have integrated multi-level acoustic features within lightweight and effective models. This study proposes a one-dimensional DCapsNet-based framework to address these limitations. The proposed framework combines spectral, temporal, and prosodic features to better capture the hierarchical relationship between speech characteristics and emotional states.

3. Proposed methodology

A classical process for SER involves three main stages. First, the extraction of emotion-specific features from speech signals. Next, the development of a model to recognize these emotions based on the features. Finally, the evaluation stage to ensure that the model accurately predicts the emotions. A detailed explanation of these stages is provided in the following sections, as illustrated in Figure 2.

3.1. Dataset description

The proposed model is evaluated using three datasets: EMO-DB, RAVDESS, and SAVEE. These datasets contain emotional speech data collected from various sources to cover a broad spectrum of emotional expressions. The EMO-DB is a publicly available dataset designed for the performance evaluation of newly developed SER models. EMO-DB [41] comprises 535 files recorded by ten speakers (5 males and 5 females) in the German language. It includes seven emotions: anger, fear, boredom, disgust, neutral, sadness, and happiness. Each file has a sample rate (SR) of 16 kHz, and the file format is waveform. Table 1 presents a summary of the EMO-DB.

The second benchmark dataset utilized in this study is RAVDESS [42]. RAVDESS is a widely employed dataset in the field of SER containing 1440 audios belonging to 24 English speakers (12 M: 12 F). The audios are classified into 8 emotions (CALM, ANGER, DISGUST, NEUTRAL, FEAR, SAD, SURPRISE, and HAPPY as given in Table 2. The SR of audio is 48 kHz and a 16-bit per sample resolution.

The third dataset utilized in this study is the SAVEE [43], which is recorded by four English students and researchers. 120 sentences have been uttered by every male speaker expressing 7 emotions: ANGER, DISGUST, FEAR, HAPPY, NEUTRAL, SAD, and SURPRISE as provided in Table 3.

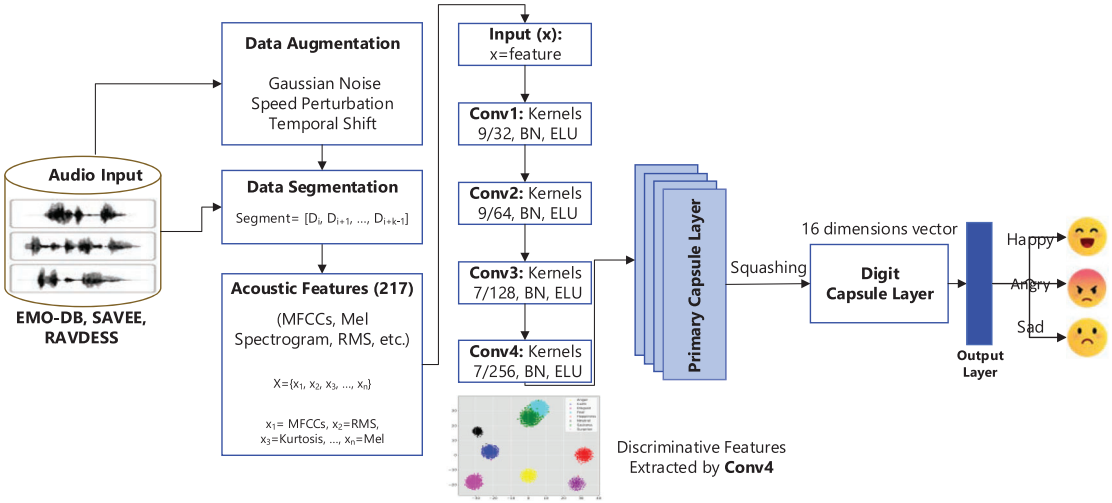


Figure 2. Proposed methodology for SER.

Table 1. Information of the EMO-DB dataset.

Emotion	Files	Percent
ANGER	127	23.73%
FEAR	69	12.89%
BOREDOM	81	15.14%
DISGUST	46	08.59%
NEUTRAL	79	14.76%
SAD	62	11.59%
HAPPY	71	13.27%
TOTAL	**535**	**100%**

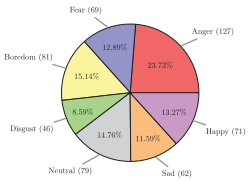


Table 2. Information of the RAVDESS dataset.

Emotion	Files	Percent
CALM	192	13.33%
ANGER	192	13.33%
DISGUST	192	13.33%
NEUTRAL	96	6.66%
FEAR	192	13.33%
SAD	192	13.33%
SURPRISE	192	13.33%
HAPPY	192	13.33%
TOTAL	**1440**	**100%**

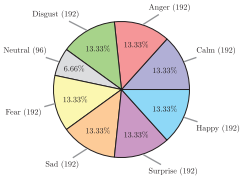
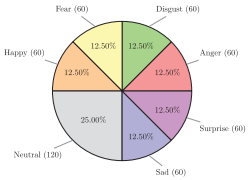


Table 3. Information of the SAVEE dataset.

Emotion	Files	Percent
ANGER	60	12.50%
DISGUST	60	12.50%
FEAR	60	12.50%
HAPPY	60	12.50%
NEUTRAL	120	25.00%
SAD	60	12.50%
SURPRISE	60	12.50%
TOTAL	**480**	**100%**



3.2. Data preprocessing

The audio files in each dataset are divided into segments using a specified window size. These segments are used as input to extract crucial information about various emotions. The segmentation techniques include both adaptive and fixed window sizes. In this study, a fixed window size of 128 samples is used for segmentation, with an overlapping ratio of 50% to obtain a collection of segments that are used to retrieve temporal and spectral features. Specifically, the set of segments is denoted as $S = \{S_1, S_2, \ldots, S_N\}$, where N represents the total number of segments. Each segment is defined as a sequence of length K, as shown in Equation (1).

$$\text{Segment} = [D_i, D_{i+1}, , D_{i+K-1}] \tag{1}$$

where, D_i represents the data at the ith position.

3.3. Data augmentation

Data augmentation is widely utilized in SER models to improve robustness, support generalization, and mitigate overfitting. Although emotions are highly sensitive to prosodic variations, the data enhancement are applied in SER due to:

(1) Emotions in speech are conveyed through slight variations in pitch, speed, and tone. These techniques help capture a wider range of speech variations and enable the model to become more adaptable, and generalized to unseen data.
(2) It introduce variability in the training data which helps to mitigate overfitting problems. An SER model trained on small datasets may struggle to perform well in real-world scenarios. By using diverse modifications, the model learns more meaningful features and avoids overfitting.
(3) Data augmentation enhance dataset diversity without additional labelled samples. With a variety of speech data, these methods enable the SER model to better recognize different emotions.

Data augmentation not only improves accuracy but also stabilizes the data distribution and reduce variance [44]. Since DL models need large-scale training

data, researchers often used the augmentation techniques like Gaussian noise addition [45], speed perturbation [46], and temporal shifting. Speed perturbation alters the speed and duration of an audio signal by resampling it at a randomly chosen rate, and introduces temporal variations that enhance dataset diversity. This augmentation technique helps the model learn robust features by exposing it to varied speech tempos. For each sample, a stretching rate uniformly from [0.9, 1.1] is implemented. This speeds up or slows down the audio by up to 10% on every augmentation call. The resultant signal is truncated or zero-padded back to the original length. However, excessive time stretching may introduce distortions and can affect data quality. Temporal shift moves the waveform forward or backward by a random offset. The gap left behind is filled with silence. This mimics real-world timing issues, like a speaker starting late or a mic lagging behind. It helps the model handle small timing mismatches between the emotional cue and the start of the recording. A random shift magnitude up to 10% (i.e. up to $0.10 \times sr$ samples) is implemented in this study. However, excessive temporal shifting could shift meaningful content out of the analysis window. Therefore, the shift magnitude is kept small. To improve model robustness against noisy data, Gaussian noise is added to the original audio. Nonetheless, the selection of an optimal value for the hyperparameter "σ" to control noise amplitude is critical. Excessive noise (high σ) can hinder optimization, while insufficient noise (low σ) may fail to enhance generalization [47]. Gaussian noise (noise factor = 0.001) improves robustness by training the model on varied inputs. Moreover, it simulates realistic environmental variations which makes it adaptable to diverse recording environments. Each augmentation technique increase the data diversity and improve the model generalization. However, careful parameter selection is important to avoid compromised data quality. Each augmentation operation can be formally defined as a transformation function $T_j : \mathbb{R}^n \rightarrow \mathbb{R}^n$ which is applied to each audio sample x_i:

$$x_i' = T_j(x_i) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \tag{2}$$

where T_j represents speed perturbation, temporal shift, or Gaussian noise addition. The augmented dataset becomes $D' = \{(x_i', y_i)\}_{i=1}^{N'}$ and ensure $N' > N$ to enhance generalization.

Data augmentation was applied in a leakage-safe manner within the cross-validation procedure. Specifically, augmentation was applied exclusively to the training partition after the train/validation split for each fold. The validation fold in every iteration contained only original, unaugmented samples. This ensures that no augmented derivative of a validation sample was present in the training set.

3.4. Feature extraction

The mining of distinct and useful information from an audio file is crucial to accurately classify speech emotions. Recently, several authors have recommended various features for SER, such as MFCCs and perceptual linear predictive (PLP). These features have gained popularity due to the ability to capture recurring patterns and overall effectiveness in speech signals. The MFCC features have shown effectiveness in numerous experiments for SER. However, the performance of these features degrades in a complex collection of speech signals, making it difficult to retrieve specific emotion features. To address this issue, this study proposed a method that integrates both spectral and temporal features. Spectral features capture the distribution of energy within the speech signal. Conversely, the temporal features involve extracting statistical metrics from speech signals to provide valuable insights into the temporal progression of the speech signal. These features are valuable for real-time emotion recognition due to their efficiency and minimal computational demands. Therefore, it is crucial to use window sizes of d_t seconds, represented by $N = f_s \times d_t$ samples, to capture a range of feature vectors. Here, f_s stands for the sampling frequency. The process of combining these extracted features is explained below:

MFCC features are employed in the field of signal processing and SER. To obtain MFCC features, the audio file is initially segmented into fixed-length frames. Secondly, a technique called windowing is applied to reduce silence across all frames of the speech file. To capture key characteristics across both low and high resonance frequencies, digital filters are utilized in both linear and logarithmic forms.

$$Mel_{freq}(f) = 2595 \times \log_{10}\left(1 + \frac{f}{700}\right) \tag{3}$$

Spectral frequency (f) of the signal is converted to a logarithmic scale known as the "Mel", which is influenced by the pitch of the emotional signal. This "Mel" scale is used to analyze all values derived from the transformed speech signal through the Fast Fourier Transform. Following this, the power logs are calculated for each Mel-frequency and then transformed back into a time series using the Discrete Cosine Transform. The final amplitudes obtained from this spectrum are referred to as MFCCs.

The Mel-spectrogram is a key feature as it effectively captures the frequency patterns associated with speech emotions. It offers a compact representation of speech data and enhances the accuracy of SER models. Beyond spectral features, statistical metrics such as spectral kurtosis are also important. Spectral Kurtosis is used to assess the presence of outliers in the speech signal. Kurtosis measures how much a distribution deviates from a normal distribution, reflecting its

flatness or extremeness.

$$S_K = \frac{\sum_{m=1}^{R} \frac{(T_m - \bar{T})^4}{R}}{\sigma^4} \tag{4}$$

T represents the mean of the audio file, while σ indicates the standard deviation of T. S_K denotes the ratio of the fourth-order moment to the square of the second-order moment.

$$S_K = \begin{cases} < 0; & \text{Flat} \\ = 0; & \text{Normal} \\ > 0; & \text{Pleaked} \end{cases} \tag{5}$$

The degree of variation is quantitatively characterized by the skewness of its probability distribution relative to its mean. A probability distribution of a random variable that is not symmetric about its mean is formally referred to as skewed.

$$S_K = \frac{\sum_{m=0}^{R} \frac{(T_m - \bar{T})^3}{R}}{\sigma^3} \tag{6}$$

Here, R denotes the total number of emotion files.

$$S_s = \begin{cases} < 0; & \text{(-ve)ly skewed} \\ = 0; & \text{symmetric value} \\ > 0; & \text{(+ve)ly skewed} \end{cases} \tag{7}$$

Root Mean Energy (RME) feature captured the energy variations in an audio file. It measured the short-term energy of a signal and helped in distinguishing different emotions based on loudness and intensity. For instance, anger and happiness tend to have higher energy levels, whereas sadness and boredom exhibit lower energy levels. The RME of a signal X with N samples is computed as:

$$\text{RME} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2} \tag{8}$$

Chroma features represent the energy distribution of a signal across the twelve pitch, obtained by projecting the short-time Fourier transform spectrum onto these twelve chroma bins. Since pitch-class content is closely tied to intonation and prosodic variation, chroma features help capture the tonal characteristics of speech that vary with emotional expression, independent of the absolute pitch register of the speaker. The Tonnetz features project chroma information onto a six-dimensional space representing harmonic relationships such as perfect fifths and major/minor thirds. Although originally developed for harmonic analysis in music, Tonnetz features are useful in SER as they capture subtle tonal and pitch-interval relationships in the speech signal that correlate with prosodic emotional cues. Moreover, the Zero-Crossing Rate (ZCR) measures the rate at which a signal changes sign. ZCR is a simple yet effective temporal feature for distinguishing voiced from unvoiced speech segments and reflects the noisiness or turbulence of an audio signal, both of which vary systematically with emotional state. Furthermore, the spectral contrast measures the difference in amplitude between spectral peaks and valleys within multiple sub-bands of the frequency spectrum.

Table 4. Feature extraction parameters.

Feature	Parameters used
MFCCs	n_fft = 2048, hop_length = 512, sr = 22050, window = 'hann', n_mfcc = 60
Chroma	n_fft = 2048, hop_length = 512, sr = 22050, window = 'hann', n_chroma = 12
Mel	n_fft = 2048, hop_length = 512, sr = 22050, window = 'hann', n_mels = 128
ZCR	frame_length = 2048, hop_length = 512
RMS	frame_length = 2048, hop_length = 512
Tonnetz	sr = 22050, computed from harmonic component of y via chroma
Kurtosis	Applied directly to raw waveform. No additional parameters
Skewness	Applied directly to raw waveform. No additional parameters
Contrast	n_fft = 2048, hop_length = 512, sr = 22050, n_bands = 6, fmin = 200.0

Here N is the number of samples in a given frame, and x_i represents the amplitude of the signal at the ith sample.

In this study, 60 MFCC coefficients are extracted using a 2048-sample FFT window, 512-sample hop length, and 22,050 Hz sampling rate. Mel-spectrogram features are computed with identical parameters and 128 Mel bands. Kurtosis and skewness are calculated directly from the waveform, and RMS energy is computed frame-wise and averaged over time as given in Table 4. The above features are utilized to create a thorough representation of speech emotions. This fusion derived various dimensions of the input data, including spectral properties, statistical metrics, and frequency-related information. By employing multimodal information combination, the (Algorithm 1) improved the model's ability to differentiate between different emotions more accurately. Additionally, the fusion of a diverse set of features enhanced the model's generalization and robustness.

Let the extracted feature vector for each audio sample be represented as:

$$\mathbf{F}_i = [\mathbf{M}_i, \mathbf{C}_i, \mathbf{L}_i, \mathbf{Z}_i, \mathbf{T}_i, \mathbf{K}_i, \mathbf{S}_i, \mathbf{E}_i, \mathbf{N}_i] \in \mathbb{R}^d \tag{9}$$

where $\mathbf{M}_i$ denotes MFCC features, $\mathbf{C}_i$ denotes Chroma (STFT), $\mathbf{L}_i$ the Mel-spectrogram features, $\mathbf{Z}_i$ is the Zero-Cross Rate, $\mathbf{T}_i$ is the Tonnetz, $\mathbf{K}_i$ and $\mathbf{S}_i$ represent statistical metrics (kurtosis and skewness), $\mathbf{T}_i$ is the spectral contrast, and $\mathbf{E}_i$ corresponds to energy-based measures. The objective of feature extraction is to obtain a transformation $\phi : X \rightarrow \mathbf{F}$ that maximizes inter-class variance (S_B) and minimizes intra-class variance (S_W):

$$J(\phi) = \frac{\text{Tr}(S_B)}{\text{Tr}(S_W)} \tag{10}$$

Algorithm 1 Feature Extraction with Data Augmentation

Input: Audio dataset $D = \{(X_1, y_1), (X_2, y_2), \ldots, (X_N, y_N)\}$
Output: Feature vectors $F = \{\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_M\}$
Initialize F as an empty list
for each (X_i, y_i) in D **do**
  Apply Augmentation:
    $Y \leftarrow$ Gaussian Noise Injection(X_i, noise_factor = 0.001)
    $Z \leftarrow$ Speed Perturbation(X_i, rate $\sim \mathcal{U}(0.9, 1.1)$)
    $A \leftarrow$ Spectrogram Shift(X_i, shift $\leq 0.10 \times sr$)
  Feature Extraction:
  for each signal *audio* $\in \{X_i, Y, Z, A\}$ **do**
    $\mathbf{M} \leftarrow$ MFCC(*audio*, *n_mfcc* = 60)
    $\mathbf{C} \leftarrow$ Chroma-STFT(*audio*, *n_chroma* = 12)
    $\mathbf{L} \leftarrow$ Mel Spectrogram(*audio*, *n_mels* = 128)
    $\mathbf{E} \leftarrow$ Root Mean Square Energy(*audio*, $n = 1$)
    $\mathbf{Z} \leftarrow$ Zero-Crossing Rate(*audio*, $n = 1$)
    $\mathbf{T} \leftarrow$ Tonnetz(*audio*, $n = 6$)
    $\mathbf{N} \leftarrow$ Contrast(*audio*, $n = 7$)
    $\mathbf{K} \leftarrow$ Kurtosis(*audio*, $n = 1$)
    $\mathbf{S} \leftarrow$ Skewness(*audio*, $n = 1$)
    features $\leftarrow$ Concatenate($\mathbf{M}, \mathbf{C}, \mathbf{L}, \mathbf{Z}, \mathbf{T}, \mathbf{K}, \mathbf{S}, \mathbf{E}, \mathbf{N}$)
    $\mathbf{f} \leftarrow$ (**features**, y_i)
    Append $\mathbf{f}$ to F
  end for
end for

Table 5. Parameters of the DCapsNet for SER.

Layer (type)	Output shape	Param #
InputLayer	(None, 217)	0
reshape (Reshape)	(None, 217, 1)	0
conv1 (Conv1D)	(None, 209, 32)	320
batch_normalization	(None, 209, 32)	128
dropout (Dropout)	(None, 209, 32)	0
conv2 (Conv1D)	(None, 201, 64)	18,496
batch_normalization_1	(None, 201, 64)	256
dropout_1 (Dropout)	(None, 201, 64)	0
conv3 (Conv1D)	(None, 195, 128)	57,472
batch_normalization_2	(None, 195, 128)	512
dropout_2 (Dropout)	(None, 195, 128)	0
conv4 (Conv1D)	(None, 95, 256)	229,632
batch_normalization_3	(None, 95, 256)	1,024
primary_caps	(None, 736, 8)	163,968
digit_caps	(None, 8, 16)	753,664
capsule_lengths	(None, 8)	0

3.5. Deep capsule network for SER

The DCapsNet architecture adapted for SER consists of three sequential phases: a convolutional feature extraction phase, a capsule network phase, and a classification layer, as outlined in Table 5. Each phase is designed to model hierarchical acoustic patterns and spatial relationships in speech signals.

The convolutional phase employed four 1D convolutional (Conv) layers to extract discriminative temporal and spectral information from acoustic representations. Each layer performed convolution, batch normalization (BN), an ELU activation, and a dropout of 0.1 for regularization. The first two Conv layers used a kernel size of 9, the third and final layer used a kernel size of 7. A stride of 1 is used in the first three convolutional layers to preserve fine-grained temporal resolution, while the final layer (Conv4) uses a stride of 2 to down-sample the temporal dimension before the capsule stage. Given an input speech signal $X = [x_1, x_2, \ldots, x_T] \in \mathbb{R}^{1\times T}$, where T is the signal length. Each Conv layer computes feature maps $H_j^l \in \mathbb{R}^{K^l \times D^l}$ through:

$$H_j^l = \sum_{i=1}^{K^{l-1}} H_i^{l-1} * w_{ji}^l + b_j^l, \tag{11}$$

where w_{ji}^l denotes the kernel weights and b_j^l the bias term.

To stabilize training and accelerate convergence, BN is utilized to normalize the feature map outputs within each batch. The BN transformation is given by:

$$\hat{H}_j^l = \gamma_c \frac{H_j^l - E[H_j^l]}{\sqrt{Var[H_j^l] + \epsilon}} + \beta_c, \tag{12}$$

where γ_c and β_c are learnable parameters, and $\epsilon > 0$ ensures numerical stability. The ELU function introduces nonlinearity while mitigating vanishing gradients:

$$f(\hat{H}_j^l) = \begin{cases} \hat{H}_j^l & \text{if } \hat{H}_j^l \geq 0, \\ e^{\hat{H}_j^l} - 1 & \text{otherwise.} \end{cases} \tag{13}$$

At the end, each Conv layer employed an Exponential Linear Unit (ELU) activation function to transform feature maps, represented as $H_j^{conv_l} = f(\hat{H}_j^l)$. The ELU function is:

$$f(\hat{H}_j^l) = \begin{cases} H_j^l, & \text{for } H_j^l \geq 0 \\ \left(e^{H_j^l} - 1\right), & \text{for } H_j^l < 0 \end{cases} \tag{14}$$

The well-formed clusters in Figure 3(a,b) show that the DCapsNet effectively learned distinct feature representations for different emotions to help the improved classification results. The improved separation of clusters in this visualization demonstrates strong feature learning, with minimal overlap between clusters, indicating similarities between certain emotions.

The capsule network phase in DCapsNet is specifically designed to model hierarchical relationships in speech features beyond what conventional CNNs capture. Following the convolutional feature extraction stage, the 256-channel feature maps produced by the final Conv layer are transformed into vector-based capsules. The primary capsule layer consists of 736 capsules, each with an 8-dimensional vector output, obtained by applying a 1D convolutional projection (128 filters, kernel size 5, stride 2, "same" padding) to the final convolutional feature maps H_{conv_L}, followed by a ReLU pre-activation (no squash is applied at this

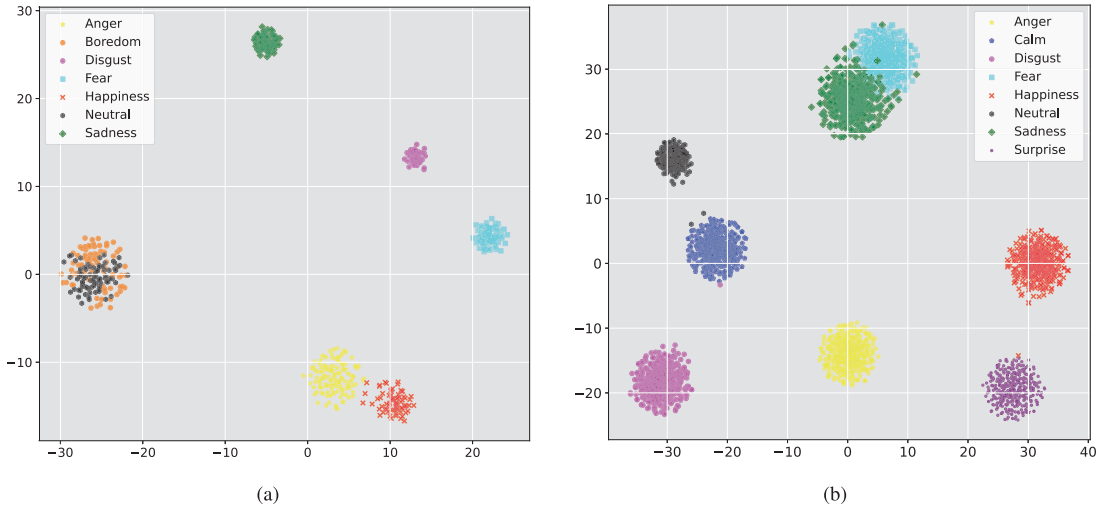


Figure 3. t-SNE Visualization of DCapsNet Features for (a) EMO-DB; (b) RAVDESS.

stage). The primary capsule output u_j for capsule j is therefore computed as:

$$u_j = \text{ReLU}\left(\text{Conv1D}_{5,2}(H_{conv_L})\right)_j. \tag{15}$$

The higher-level (digit) capsule layer then aggregates the 736 primary capsules into 8 digit capsules – one per emotion class – each represented by a 16-dimensional vector v_j, via dynamic routing-by-agreement. Within each routing iteration, the weighted sum s_j of the routed primary-capsule predictions is passed through the Squash non-linearity to obtain v_j:

$$v_j = \text{Squash}(s_j) = \frac{\|s_j\|^2}{1+\|s_j\|^2}\frac{s_j}{\|s_j\|}. \tag{16}$$

A final Length layer computes the L_2-norm of each digit capsule vector v_j, yielding a score for each of the 8 emotion classes. For routing, this study employed a dynamic routing-by-agreement mechanism with three routing iterations, where routing coefficients c_{ij} between capsule i in the primary layer and capsule j in the next layer are updated as:

$$c_{ij} = \frac{\exp(b_{ij})}{\sum_k \exp(b_{ik})}, \quad b_{ij} \leftarrow b_{ij} + u_i \cdot v_j, \tag{17}$$

where b_{ij} are routing logits and v_j is the output vector of capsule j. This iterative process encourages capsules to send their outputs to parent capsules with which they agree, thereby capturing part-whole relationships in emotional speech patterns. Unlike a conventional softmax classifier, no additional fully connected layer follows the capsule stage – the digit capsule norms $\|v_j\|$ themselves are used directly as the predicted class probabilities and are optimized via the margin loss described below. The complete architecture diagram is presented in Figure 4.

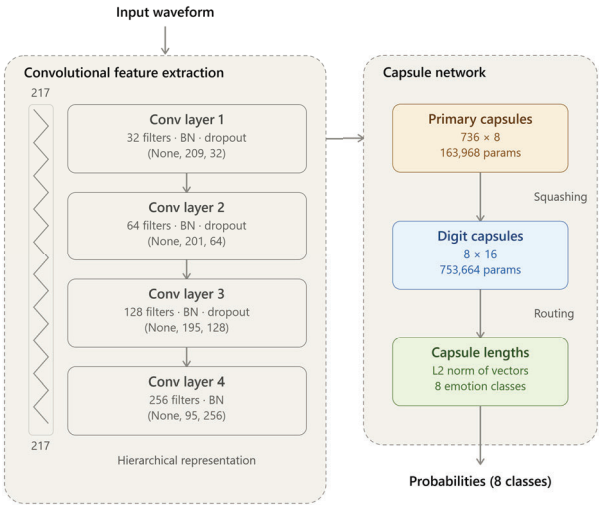


Figure 4. Proposed DCapsNet architecture for SER.

The network parameters $\Theta = \{W\}$ are optimized by minimizing the categorical margin loss $\mathcal{L}_{\text{margin}}$:

$$\mathcal{L}_{\text{margin}} = \sum_{i=1}^{N} T_i \max(0, m^+ - \|v_i\|)^2 + \lambda(1-T_i)\max(0, \|v_i\| - m^-)^2 \tag{18}$$

where $T_i = 1$ if emotion i is present, and $T_i = 0$ otherwise. The parameters $m^+ = 0.9$, $m^- = 0.1$, and $\lambda = 0.5$ control the margin boundaries. This implementation trains on the margin loss alone; no reconstruction decoder or reconstruction loss term is used.

3.6. Evaluation metrics

The performance of the deep capsule network was evaluated by assessing its capability to accurately recognize emotions. This study employed the metrics specified in this section for evaluation. To compute the evaluation metrics for various methods, the confusion matrix (CM) of each individual classifier is utilized. Accuracy,

precision, recall, F1-score, and root mean squared error were utilized to evaluate the performance of DCapsNet for SER by using the CM. The CM consists of TN (true-negative), TP (true-positive) when the SER model correctly recognizes emotions, and FN (false-negative), FP (false-positive) when the SER model incorrectly recognizes. The classification accuracy is determined by calculating the percentage of correctly identified emotions, which is the sum of successfully recognized emotions, divided by the total number of emotion audio files analyzed using Equation (19).

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{TP+TN}{TP+FP+TN+FN}\right)_i \quad (19)$$

where N shows the total number of audios.

Precision is a widely utilized metric to evaluate the efficacy of the SER method, especially when the dataset comprises a significant instance of FP, which can impact average precision. It is calculated by the proportion of correctly recognized positive emotion audio files to the total positive emotions, as shown in Equation (20).

$$\text{Precision} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{TP}{TP+FP}\right)_i \quad (20)$$

Recall is computed by the proportion of correctly recognized positive emotional audios to the total of correctly recognized positive audios and incorrectly recognized negative audios, as given in Equation (21).

$$\text{Recall} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{TP}{TP+FN}\right)_i \quad (21)$$

F1-score measure is commonly used to reduce the trade-offs between accuracy and recall. It is calculated by taking the average precision and recall as shown in Equation (22).

$$\text{F1-score} = \frac{1}{N}\sum_{i=1}^{N} 2\times\left(\frac{\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}\right)_i \quad (22)$$

Root Mean Squared Error (RMSE) is another commonly used metric to evaluate the performance of SER models. The RMSE is a quadratic assessment measure that quantifies the average number of errors classifiers incur. It computes the difference between predicted and actual emotions. A lower RMSE value indicates better model performance. Mathematically, it is written as given in Equation (23).

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i-\hat{y}_i)^2} \quad (23)$$

Here N is the total number of audios, y_i is the actual emotion value for audio i, and $\hat{y}_i$ is the predicted emotion value for audio i.

4. Experiments

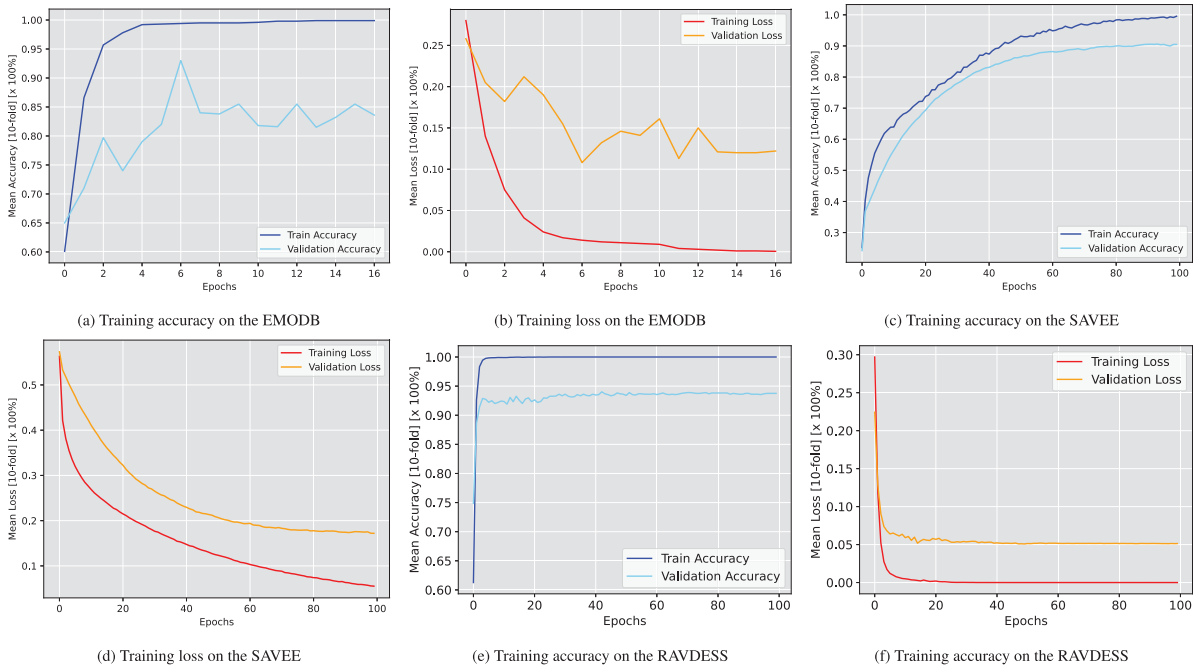
To ensure robust and unbiased evaluation, this study adopted a 10-fold cross-validation strategy on each dataset. The datasets were first partitioned into 10 equally sized folds while preserving the original class distribution in each fold. In every iteration, nine folds were used for training and one fold for testing to ensure that each sample appeared exactly once in the test set. Crucially, data augmentation was applied only to the training partition of each fold, after the train/validation split was determined. For training, we used the Adam optimizer with a learning rate of 0.0001, a batch size of 8, and a maximum of 100 epochs. Early stopping with a patience of 10 epochs was applied to avoid overfitting. The experiments were conducted on a MacBook Pro (13-inch, Mid 2012) equipped with a 2.5 GHz dual-core Intel Core i5 processor, 8 GB of RAM, a 500 GB hard drive, and Intel HD Graphics 4000. The software environment consisted of macOS, Python 3.9, TensorFlow/Keras (v 2.19.0), NumPy, scikit-learn, and Matplotlib.

Figure 5 summarizes the training performance of the DCapsNet models across the datasets. The proposed framework demonstrated exceptional results and achieved nearly perfect accuracy (99–100%) on all datasets during training. This highlights the robustness of the proposed SER framework in minimizing training-testing loss while optimizing classification efficacy. The training accuracy plots exhibit a rapid convergence towards 99–100% accuracy within the early epochs, which indicates the stable and effective learning. Similarly, the corresponding loss curves show a sharp decline to confirm the efficient optimization and minimal overfitting. The consistency of these patterns across all datasets validates the robustness of the proposed DCapsNet framework. The smooth and stable nature of the curves also indicates that the model successfully mitigates vanishing gradient issues to ensure effective backpropagation.

4.1. DCapsNet performance

This study utilized three datasets–EMO-DB, SAVEE, and RAVDESS– to evaluate the performance of the proposed DCapsNet for SER. The classification reports of the DCapsNet on each dataset are presented in Tables 6–8, which demonstrates the robustness of the DCapsNet model compared to existing baseline models [48, 49].

The results indicate that the SER model performed significantly well on the EMO-DB dataset and achieved an average accuracy of 93%. Notably, DCapsNet outperformed previous models across all emotions in EMO-DB, including the challenging emotion of boredom. However, the model showed comparatively lower accuracy for disgust, as it exhibited confusion with other emotions. This misclassification is visible in Figure 6(a),



(a) Training accuracy on the EMODB (b) Training loss on the EMODB (c) Training accuracy on the SAVEE

(d) Training loss on the SAVEE (e) Training accuracy on the RAVDESS (f) Training accuracy on the RAVDESS

Figure 5. Training & validation performance of DCapsNet across different datasets. (a) Training accuracy on the EMODB. (b) Training loss on the EMODB. (c) Training accuracy on the SAVEE. (d) Training loss on the SAVEE. (e) Training accuracy on the RAVDESS. (f) Training accuracy on the RAVDESS.

Table 6. Mean 10-fold cross-validation performance of DCapsNet on the EMODB dataset.

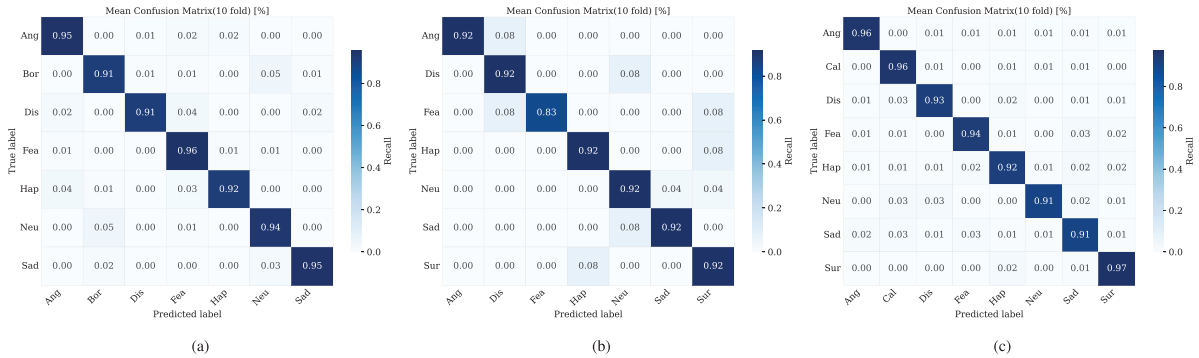
Class	Accuracy	Precision	Recall	F1-Score	RMSE	G-Mean	Std Dev	95% CI
anger	0.9703	0.9603	0.9528	0.9565	0.1434	0.9701	0.1435	(0.9159, 0.9897)
boredom	0.9502	0.9250	0.9136	0.9193	0.1559	0.9495	0.1560	(0.8524, 0.9748)
disgust	0.9545	0.9545	0.9130	0.9333	0.1059	0.9536	0.1059	(0.8316, 0.9945)
fear	0.9697	0.8919	0.9565	0.9231	0.1434	0.9696	0.1432	(0.9084, 1.0000)
happiness	0.9534	0.9420	0.9155	0.9286	0.1367	0.9527	0.1368	(0.8508, 0.9802)
neutral	0.9607	0.9136	0.9367	0.9250	0.1498	0.9604	0.1499	(0.8830, 0.9904)
sadness	0.9737	0.9672	0.9516	0.9593	0.0967	0.9734	0.0967	(0.8982, 1.0000)
Micro	0.9364	0.9364	0.9342	0.9350	0.1331	0.9613	0.1332	(0.8772, 0.9899)

Table 7. Mean 10-fold cross-validation performance of DCapsNet on the SAVEE dataset.

Class	Accuracy	Precision	Recall	F1-Score	RMSE	G-Mean	Std Dev	95% CI
anger	0.9583	1.0000	0.9167	0.9565	0.1021	0.9574	0.1021	(0.7603, 1.0000)
disgust	0.9464	0.8462	0.9167	0.8800	0.1768	0.9460	0.1774	(0.7603, 1.0000)
fear	0.9167	1.0000	0.8333	0.9091	0.1443	0.9129	0.1436	(0.6225, 1.0000)
happiness	0.9524	0.9167	0.9167	0.9167	0.1443	0.9517	0.1451	(0.7603, 1.0000)
neutral	0.9444	0.9167	0.9167	0.9167	0.2041	0.9440	0.2052	(0.8061, 1.0000)
sadness	0.9524	0.9167	0.9167	0.9167	0.1443	0.9517	0.1451	(0.7603, 1.0000)
surprise	0.9405	0.7857	0.9167	0.8462	0.2041	0.9402	0.2041	(0.7603, 1.0000)
Micro	0.9062	0.9117	0.9048	0.9060	0.1600	0.9434	0.1604	(0.7471, 1.0000)

Table 8. Mean 10-fold cross-validation performance of DCapsNet on the RAVDESS dataset.

Class	Accuracy	Precision	Recall	F1-Score	RMSE	G-Mean	Std Dev	95% CI
anger	0.9760	0.9583	0.9583	0.9583	0.1054	0.9758	0.1054	(0.9383, 0.9783)
calm	0.9732	0.9246	0.9583	0.9412	0.1264	0.9730	0.1263	(0.9383, 0.9783)
disgust	0.9599	0.9519	0.9271	0.9393	0.1264	0.9594	0.1264	(0.9011, 0.9531)
fear	0.9651	0.9524	0.9375	0.9449	0.1208	0.9647	0.1208	(0.9133, 0.9617)
happiness	0.9561	0.9365	0.9219	0.9291	0.1369	0.9555	0.1369	(0.8950, 0.9487)
neutral	0.9509	0.9355	0.9062	0.9206	0.1021	0.9498	0.1021	(0.8650, 0.9475)
sadness	0.9459	0.9062	0.9062	0.9062	0.1581	0.9451	0.1581	(0.8771, 0.9354)
surprise	0.9822	0.9397	0.9740	0.9565	0.1087	0.9821	0.1086	(0.9580, 0.9899)
Micro	0.9382	0.9381	0.9362	0.9370	0.1231	0.9632	0.1231	(0.9108, 0.9616)



(a) (b) (c)

Figure 6. Confusion matrices for DCapsNet on (a) EMO-DB, (b) SAVEE, and (c) RAVDESS. The y-axes shows the actual emotions and the x-axis shows the predicted emotions.

Table 9. Computational efficiency analysis of the proposed DCapsNet model on three datasets.

Metric	Unit	EMO-DB	SAVEE	RAVDESS
Iteration time	seconds/iteration	0.0250	0.0241	0.0492
Total training time	seconds	978.75	880.87	2599.41
Inference latency	ms/sample	4.062	3.936	4.127
Throughput	samples/s	246.21	254.04	242.28
Model parameters	M	1.225	1.225	1.226
Approx. FLOPs	GFLOPs	0.015	0.015	0.015
Model size	MB	7.08	7.07	7.10
Peak memory usage	MB	586.30	799.58	661.75

where correctly predicted emotions appear along the diagonal, while inter-emotion misclassifications are shown in the corresponding rows.

For the SAVEE dataset, the proposed model demonstrated low recall for fear (83%) emotion as illustrated in Figure 6(b). However, disgust emotion achieved a slightly higher recall of 92% compared to other emotions. Similarly, the RAVDESS dataset achieved notable recall for the emotions of surprise (97%), anger (96%), and calm (96%), as shown in Figure 6(c). Additionally, fear emotion also achieved superior recall compared to the baseline RAVDESS SER model [48, 49]. Overall, the recall of DCapsNet across RAVDESS outperformed the existing models. Overall, the proposed DCapsNet-based SER model achieved high prediction accuracy across all datasets, reduced model size, and improved computational efficiency.

4.2. Computational efficiency

Table 9 presents a comprehensive analysis of the computational efficiency of the DCapsNet model on three datasets: EMO-DB, SAVEE, and RAVDESS. All experiments were conducted on a MacBook Pro (2.5 GHz Intel Core i5, 8 GB RAM, Intel HD 4000 GPU), a typical consumer-grade system. This demonstrates the model's ability to perform well on modest hardware. Despite these constraints, DCapsNet shows good efficiency across several key performance metrics. The iteration times are low, from 0.0241 s to 0.0492 s per iteration which indicates the model's responsiveness and fast convergence. Training times vary slightly between datasets, with RAVDESS taking the longest at 2599.41 s, but all remain highly efficient compared to similar DL models. Notably, inference latency stays consistently under 5 milliseconds per sample which highlights the model's real-time potential for emotion recognition applications. The model achieves high throughput without sacrificing recognition accuracy or processing speed.

With approximately 1.23 million parameters and a model size of less than 7.1 MB, DCapsNet strikes an optimal balance between compactness and performance. It is well-suited for deployment on resource-constrained devices such as mobile platforms and edge devices. Peak memory usage stays well below 800 MB which shows that DCapsNet operates efficiently on systems with limited computational resources. These results highlight the model's computational efficiency and position it as an ideal solution for real-time emotion recognition across a wide range of devices. It achieves this without compromising performance or requiring significant hardware.

5. Discussion

The performance of the proposed DCapsNet-based SER model was evaluated against baseline methods using EMO-DB, SAVEE, and RAVDESS to assess its robustness. As shown in Table 10, the DCapsNet model outperformed baseline methods across each dataset and obtained an micro accuracy of 93.0% for EMO-DB, 90.0% for SAVEE, and 94.0% for RAVDESS. While certain existing methods have reported high performance for specific speech emotions, such as anger in RAVDESS with 95.0% accuracy [50], the performance

Table 10. Comparison of DCapsNet with baseline methods for SER.

Study	Dataset	Accuracy (%) of all emotions									
		Neutral	Calm	Happy	Sad	Angry	Fear	Disgust	Surprise	Bored	Micro
[48]	RAVDESS	0.68	0.90	0.65	0.66	0.80	0.74	0.71	0.67	×	0.73
	EMO-DB	0.90	×	0.92	0.93	0.92	0.92	0.99	×	0.88	0.90
	SAVEE	0.82	×	0.47	0.58	0.90	0.50	0.48	0.53	×	0.67
[49]	RAVDESS	0.75	0.57	0.68	0.52	0.92	0.76	0.72	0.80	×	0.71
	EMO-DB	1.00	×	1.00	0.87	1.00	0.67	0.67	×	0.61	0.86
[50]	RAVDESS	0.85	0.95	0.43	0.61	0.95	0.91	0.86	0.95	×	0.77
	EMO-DB	0.85	×	0.66	0.88	0.91	0.92	0.87	×	0.90	0.85
This	EMO-DB	0.96	×	0.95	0.97	0.97	0.97	0.95	×	0.95	**0.93**
	SAVEE	0.94	×	0.95	0.95	0.96	0.92	0.95	0.94	×	**0.90**
	RAVDESS	0.95	0.97	0.95	0.95	0.97	0.96	0.96	0.98	×	**0.94**

of DCapsNet remained superior. This improved performance is consistent with the expected benefit of DCapsNet to capture hierarchical feature relationships and preserve spatial characteristics, which are often lost in traditional pooling operations utilized in CNNs. These experimental results confirmed that the proposed method is more reliable, accurate, and efficient compared to baseline SER methods.

To compare the proposed DCapsNet with other DL models, different frequency and time-domain features were derived from audio files. These features were obtained after utilizing signal preprocessing and data augmentation methods to improve the diversity and quality of the datasets. The derived features were then fed as input to DL models: 1-D CNN, RNN, LSTM, and BiLSTM, to construct four SER models. This study conducted 12 experiments and evaluated these four SER models across each dataset. Each model was integrated with a handcrafted feature set to assess its effectiveness in emotion recognition. The comparative results show the superior performance of proposed DCapsNet over other DL models across each dataset, as illustrated in Figure 7. The proposed model achieved an average accuracy of 94.65% on SAVEE, 97.11% on RAVDESS, and 96.17% on EMO-DB. BiLSTM achieved the highest accuracy among traditional models with 93.1% on EMO-DB and outperformed 1-D CNN (91.3%), LSTM (91%), and RNN (89.2%). In contrast, the 1-D CNN model consistently achieved lower accuracy across each dataset. Overall, the results revealed that the DCapsNet delivered better performance compared to baseline DL models across each dataset. To ensure a fair comparison, we describe the baseline models and their training settings as follows.

- 1-D CNN was comprised of four convolutional layers with ReLU activation, max-pooling, and a fully connected layer. Trained for 100 epochs using Adam optimizer with a learning rate of 0.0001 and batch size of 8.
- RNN was comprised of four recurrent layers with 128 hidden units and tanh activation function, trained with the Adam optimizer and learning rate of 0.0001.

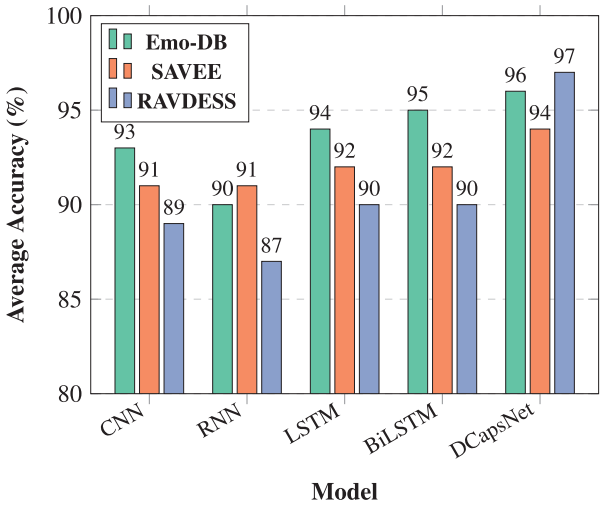


Figure 7. Comparison analysis of DCapsNet with DL models across different datasets.

- LSTM and BiLSTM both were comprised of four LSTM layers of 128 units, followed by a dense layer with softmax activation. BiLSTM used bidirectional connections. Both models were trained for 100 epochs with the Adam optimizer and a learning rate of 0.0001.

All baseline models used the same expert-driven feature set and data augmentation techniques as DCapsNet to ensure consistency in comparison.

Beyond quantitative improvements, the proposed DCapsNet offers unique strengths over other SER methods. Unlike CNNs that lose spatial information during pooling, capsule networks preserve hierarchical relationships in speech features and enable more robust emotion representation. Compared to RNNs and LSTMs, DCapsNet requires fewer parameters and avoids issues such as vanishing gradients, making it more computationally efficient for real-time applications. However, the model also presents certain limitations. The evaluation followed a sample-independent rather than a speaker-independent protocol, which may overestimate generalization performance across unseen speakers. A speaker-independent cross-validation setup will be adopted in future work to

provide a more rigorous assessment of model generalizability. It is also important to note that the reported performance reflects the combined effect of feature fusion, data augmentation, and the capsule network architecture. The individual contribution of each component has not been isolated through ablation experiments.

It should be noted that this comparison is qualitative rather than empirical, as wav2vec, HuBERT, and Transformer-based models were not re-implemented or evaluated under identical experimental conditions in this study. Based on accuracy figures reported in the literature for similar SER benchmarks, DCapsNet appears to achieve comparable recognition performance while requiring substantially fewer parameters and lower computational overhead; however, a direct, controlled comparison against these self-supervised and Transformer-based approaches is needed to substantiate this claim and is identified as a direction for future work. While Transformers are strong in modelling long-range dependencies in speech data, they generally require large-scale datasets and extensive training resources, which DCapsNet does not necessitate. These findings suggest that capsule networks, such as DCapsNet, can enhance SER in domains including call centers, assistive technologies, and emotion-aware dialogue systems. To further enhance its capabilities, future work should integrate attention mechanisms or emotion-specific loss functions to address the recognition of subtle emotions. Additionally, cross-corpus and noisy-environment evaluations, as well as combining capsules with Transformer-based modules, may enhance generalization and robustness.

6. Conclusion

SER is a challenging task due to the complexities of feature extraction and model selection. Feature extraction is a crucial phase in SER, as it enhances model performance and reduces computational complexity during both training and testing. In this research, a DCapsNet model was proposed for SER by utilizing different audio characteristics as input. The proposed DCapsNet for SER effectively captured hierarchical feature representations and preserved spatial relationships between features. The model enabled better feature extraction with a reduced number of capsules and computational time. This study evaluated the model using EMO-DB, SAVEE, and RAVDESS datasets and demonstrated its robustness in recognizing emotions. The model achieved 93% micro accuracy on EMO-DB, 90% micro accuracy on SAVEE, 94% micro accuracy on RAVDESS, and outperformed baseline approaches. A detailed analysis of confusion matrices highlighted that the model performed significantly well for most emotions. For some emotions, such as disgust and happiness, the model achieved slightly lower accuracy due to confusion between emotions. Despite this, the model outperformed existing approaches for SER across multiple datasets. In the future, a comparative study will be conducted to explore the performance of deep learning-based SER models using Urdu and Arabic datasets to enhance the cross-linguistic emotion recognition capabilities. In addition, a comprehensive ablation study will be conducted to independently evaluate the contributions of multi-feature fusion, data augmentation, and capsule networks to further strengthen the understanding of their individual impacts on SER performance.

Author contributions

CRediT: **Rashid Jahangir:** Conceptualization, Formal analysis, Methodology, Writing – original draft; **Nazik Alturki:** Funding acquisition, Investigation, Project administration, Writing – review & editing; **Faiqa Hanif:** Data curation, Formal analysis, Methodology, Resources; **Mohammed Alreshoodi:** Investigation, Resources, Visualization, Writing – review & editing

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Data availability statement

The raw audio data supporting this study are openly available in Kaggle. Preprocessed feature sets, segmentation parameters, augmentation parameters, feature extraction parameters, and model hyperparameters and training configuration are reported in full within this manuscript to enable independent reproduction of the results. The implementation code for DCapsNet is available from the corresponding author upon reasonable request.

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